

LAB EXPERIMENT – PART 1

Performance Analysis Using Type-1 Hypervisor

Proxmox VE

CPU Performance Analysis of an Ubuntu Virtual Machine using Sysbench

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1. Aim

To create and verify a virtual machine on the Proxmox VE Type-1 hypervisor, inspect its CPU, memory and disk configuration, monitor resource utilization, and measure CPU performance using Sysbench.

2. Experimental Configuration

Parameter	Configured / Observed Value
Hypervisor	Proxmox VE (Type-1)
Guest OS	Ubuntu 22.04.5 LTS
CPU allocation	1 socket, 2 cores = 2 vCPU
CPU model	QEMU Virtual CPU version 2.5+
Memory allocation	2048 MiB (2 GB)
Virtual disk	20 GB
Network bridge	vmbr0
Virtualization vendor	KVM

3. Procedure

1. **Access Proxmox VE:** Open the Proxmox VE web interface and sign in with the assigned credentials.
2. **Create the VM:** Use the Create VM wizard. Configure Ubuntu as the guest OS and allocate 2 vCPU, 2 GB RAM and a 20 GB virtual disk.
3. **Configure networking:** Attach the VM to the vmbr0 bridge using the default/VirtIO network model.
4. **Install Ubuntu:** Start the VM from the Proxmox console, install Ubuntu, restart, and log in.
5. **Verify the VM:** Run `hostnamectl`, `lscpu`, `free -h` and `df -h` to verify the operating system, CPU, memory and disk.
6. **Monitor resources:** Run `top` to observe CPU utilization, memory utilization, processes and load average.
7. **Install Sysbench:** Update the package repository and install Sysbench.
8. **Run CPU benchmark:** Execute `sysbench cpu --cpu-max-prime=20000 run` and record the performance statistics.
9. **Monitor from Proxmox:** Use the VM Summary page to observe CPU, memory, network traffic and disk I/O.
10. **Shut down:** After the measurements, shut down the Ubuntu VM using `sudo poweroff` or the Proxmox shutdown option.

4. Commands Used

```
hostnamectl
```

```
lscpu
```

```
free -h
```

```
df -h
```

```
top
```

```
sudo apt update
```

```
sudo apt install sysbench -y
```

```
sysbench --version
```

```
sysbench cpu --cpu-max-prime=20000 run
```

```
sudo poweroff
```

5. Observed Results

The supplied screenshots provide the following measurable observations from the Ubuntu VM and Proxmox VE dashboard.

Metric	Observed Result
Operating system	Ubuntu 22.04.5 LTS
Architecture	x86_64
Virtual CPUs	2 CPUs / 2 cores
CPU model	QEMU Virtual CPU version 2.5+
Hypervisor / virtualization	KVM; virtualization type shown as full
Memory	Approximately 1.9 GiB total reported by free -h; Proxmox VM configured with 2.00 GiB
Disk	20 GB virtual disk; df -h shows a 20G root filesystem
Sysbench execution time	10.0005 s
Sysbench total events	15,877
Sysbench events per second	1,587.47
Sysbench minimum latency	0.59 ms
Sysbench average latency	0.63 ms
Sysbench maximum latency	1.34 ms

6. Resource Utilization Observations

The Proxmox VE screenshots show the VM running with 2 vCPUs and 2.00 GiB of allocated memory. The CPU graph shows short periods of increased utilization during activity, while the memory graph shows usage rising to roughly 1.8 GiB. Network traffic contains a pronounced transient spike, and the Disk I/O graph shows read/write activity during the observed period.

7. Result and Conclusion

Part 1 was completed on a Proxmox VE Type-1 hypervisor. The Ubuntu virtual machine was configured with 2 vCPUs, 2 GB RAM and a 20 GB virtual disk. The VM configuration was verified using standard Linux system commands, and resource utilization was monitored both inside the guest and through the Proxmox VE management interface.

For the supplied Sysbench CPU run, the VM completed the benchmark in 10.0005 seconds and processed 15,877 events at 1,587.47 events per second, with an average latency of 0.63 ms. These values form the Type-1 baseline for comparison with the Type-2 hypervisor in the next part of the experiment.

8. Screenshots / Experimental Evidence

All screenshots supplied for Part 1 are included below in the same evidence set. The images document VM identification, CPU/memory/disk inspection, Sysbench execution, and Proxmox VE resource monitoring.

```

vm01@vm01-Standard-PC-i440FX-PIIX-1996: $ hostnamectl
Static hostname: vm01-Standard-PC-i440FX-PIIX-1996
Icon name: computer-vm
Chassis: vm
Machine ID: d5f912e1bb384f5a8bfd3a9cb6b7a534
Boot ID: f910896e0b054970a08e6fd10af74a42
Virtualization: kvm
Operating System: Ubuntu 22.04.5 LTS
Kernel: Linux 6.8.0-40-generic
Architecture: x86_64
Hardware Vendor: QEMU
Hardware Model: Standard PC i440FX + PIIX, 1996_

```

Figure 1. Ubuntu VM: hostnamectl output showing the hostname, Ubuntu operating system, kernel, architecture and virtual hardware information.

```

vm01@vm01-Standard-PC-i440FX-PIIX-1996: $ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          40 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 2
On-line CPU(s) list:    0,1
Vendor ID:              GenuineIntel
Model name:             QEMU Virtual CPU version 2.5+
CPU family:             15
Model:                  107
Thread(s) per core:     1
Core(s) per socket:     2
Socket(s):              1
Stepping:               1
BogoMIPS:               4224.00
Flags:                  fpu de pse tsc msr pae mce cx8 apic sep mtrr pge mca c
mov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx
lm constant_tsc nopl xtopology cpuid tsc_known_freq pn
i ssse3 cx16 sse4_1 sse4_2 x2apic popcnt aes hypervisor
r lahf_lm cpuid_fault ptl

Virtualization features:
Hypervisor vendor:     KVM
Virtualization type:    full
Caches (sum of all):
L1d:                    64 KiB (2 instances)
L1i:                    64 KiB (2 instances)
L2:                      8 MiB (2 instances)
L3:                      16 MiB (1 instance)
NUMA:
NUMA node(s):          1
NUMA node0 CPU(s):     0,1
Vulnerabilities:
Gather data sampling:   Not affected
Itlb multihit:          KVM: Mitigation: VMX unsupported
L1tf:                   Mitigation; PTE Inversion
Mds:                    Vulnerable: Clear CPU buffers attempted, no microcode;

```

Figure 2. Ubuntu VM: lscpu output showing x86_64 architecture, 2 CPUs/cores, QEMU Virtual CPU model and KVM virtualization information.

```

Virtualization features:
Hypervisor vendor:    KVM
Virtualization type:  full
Caches (sum of all):
L1d:                  64 KiB (2 instances)
L1i:                  64 KiB (2 instances)
L2:                   8 MiB (2 instances)
L3:                   16 MiB (1 instance)
NUMA:
NUMA node(s):         1
NUMA node0 CPU(s):    0,1
Vulnerabilities:
Gather data sampling: Not affected
Itlb multihit:        KVM: Mitigation: VMX unsupported
L1tf:                 Mitigation; PTE Inversion
Mds:                  Vulnerable: Clear CPU buffers attempted, no microcode;
                      SMT Host state unknown
Meltdown:              Mitigation; PTI
Mmio stale data:       Unknown: No mitigations
Reg file data sampling: Not affected
Retbleed:              Not affected
Spec rstack overflow:  Not affected
Spec store bypass:     Vulnerable
Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:            Mitigation; Retpolines; STIBP disabled; RSB filling; PBRSB-eIBRS Not affected; BHI Retpoline
Srbds:                 Not affected
Tsx async abort:       Not affected
vm01@vm01-Standard-PC-i440FX-PIIX-1996:~$

```

Figure 3. Ubuntu VM: extended lscpu output showing cache, NUMA and CPU vulnerability/mitigation information.

```

vm01@vm01-Standard-PC-i440FX-PIIX-1996:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi       765Mi       449Mi        13Mi        753Mi       1.0Gi
Swap:          2.1Gi          0B        2.1Gi
vm01@vm01-Standard-PC-i440FX-PIIX-1996:~$

```

Figure 4. Ubuntu VM: free -h output showing memory and swap allocation/usage.


```

vm01@vm01-Standard-PC-l440FX-PIIX-1996:~$ free -h
Mem:               total        used        free      shared  buff/cache   available
Swap:             2.1Gi          0B          2.1Gi
vm01@vm01-Standard-PC-l440FX-PIIX-1996:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           197M  1.4M  196M   1% /run
/dev/sda3       20G   9.5G   8.6G  53% /
tmpfs           985M   0  985M   0% /dev/shm
tmpfs           5.0M   0   5.0M   0% /run/lock
/dev/sda2       512M   6.1M  506M   2% /boot/efi
tmpfs           197M  112K  197M   1% /run/user/1000
/dev/sr0        4.5G   4.5G   0 100% /media/vm01/Ubuntu 22.04.5 LTS amd64
vm01@vm01-Standard-PC-l440FX-PIIX-1996:~$ top

top - 01:03:37 up 3 min,  1 user,  load average: 0.07, 0.10, 0.04
Tasks: 192 total,  1 running, 191 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.3 us,  0.2 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 1968.2 total,  448.6 free,  765.8 used,  753.8 buff/cache
MiB Swap: 2140.0 total,  2140.0 free,  0.0 used,  1044.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+    COMMAND
 398 systemd+ 20   0   14836   6784   6016 S   0.3   0.3   0:00.16 systemd-oomd
 952 vm01      20   0 4216276 345596 144436 S   0.3  17.1   0:04.20 gnome-shell
    1 root      20   0 166544   11740   8284 S   0.0   0.6   0:00.01 systemd
    2 root      20   0     0     0     0 S   0.0   0.0   0:00.00 kthreadd
    3 root      20   0     0     0     0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root      0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root      0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
    6 root      0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-slub_
    7 root      0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-netns
    8 root      20   0     0     0     0 I   0.0   0.0   0:00.00 kworker/0:0-rcu_gp
    9 root      20   0     0     0     0 I   0.0   0.0   0:00.01 kworker/0:1-events
   10 root      0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
   11 root      20   0     0     0     0 I   0.0   0.0   0:00.30 kworker/u4:0-flush-8:0
   12 root      0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-mm_pg
   13 root      20   0     0     0     0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
   14 root      20   0     0     0     0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   15 root      20   0     0     0     0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   16 root      20   0     0     0     0 S   0.0   0.0   0:00.01 ksoftirqd/0

```

Figure 5. Ubuntu VM: combined free -h, df -h and top output showing memory, filesystem and live process/resource information.

```

vm01@vm01-Standard-PC-l440FX-PIIX-1996:~$ sysbench --version
sysbench 1.0.20
vm01@vm01-Standard-PC-l440FX-PIIX-1996:~$ sysbench cpu --cpu-max-prime=20000 run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Prime numbers limit: 20000
Initializing worker threads...

Threads started!

CPU speed:
  events per second: 1587.47

General statistics:
  total time:                               10.0005s
  total number of events:                   15877

Latency (ms):
  min:                                       0.59
  avg:                                       0.63
  max:                                       1.34
  95th percentile:                         0.65
  sum:                                       9996.82

Threads fairness:
  events (avg/stddev):                   15877.0000/0.00
  execution time (avg/stddev):           9.9968/0.00
vm01@vm01-Standard-PC-l440FX-PIIX-1996:~$

```

Figure 6. Ubuntu VM: Sysbench CPU benchmark output for --cpu-max-prime=20000.

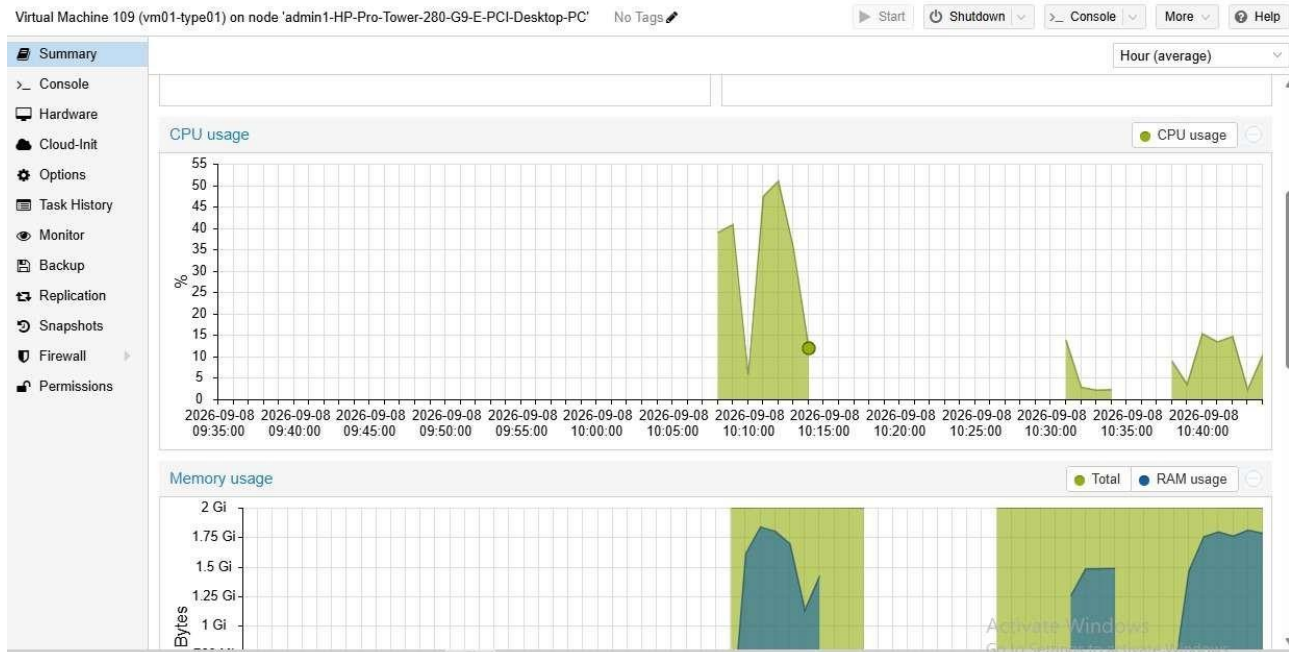


Figure 7. Proxmox VE: Summary/monitoring view showing CPU and memory usage graphs for the running VM.

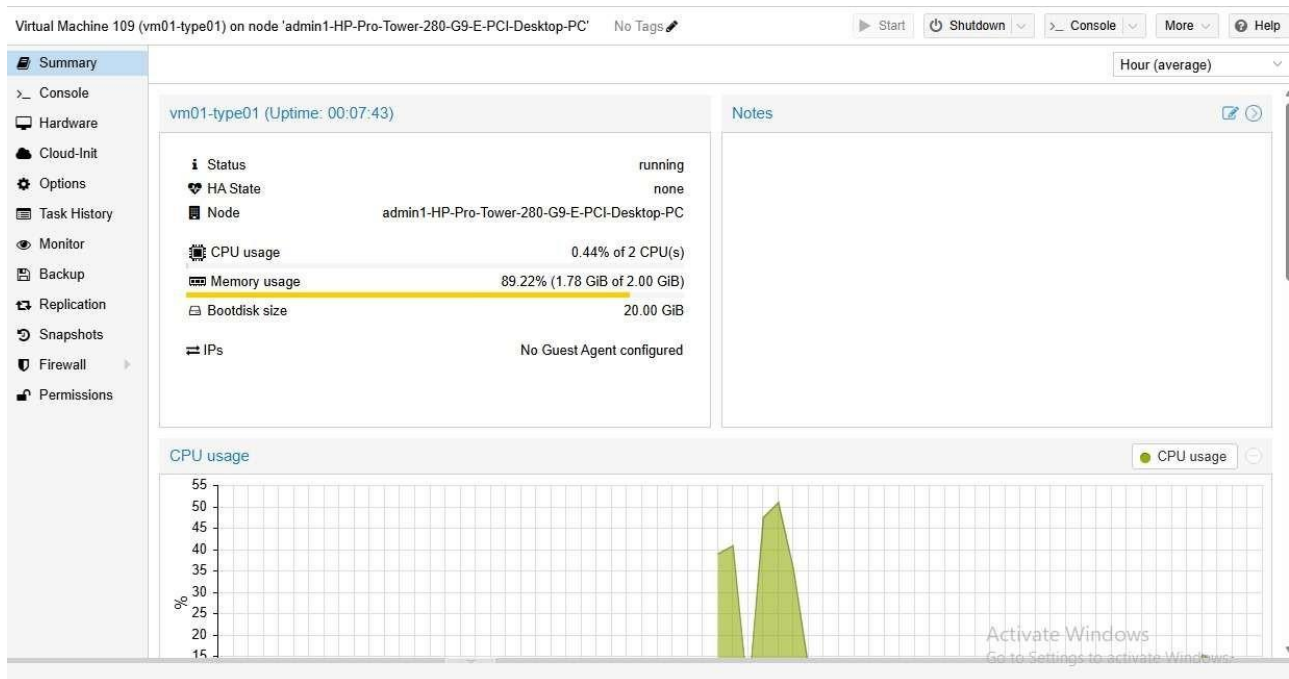


Figure 8. Proxmox VE: VM Summary showing running status, 2 CPUs, 2.00 GiB memory and 20.00 GiB boot disk.

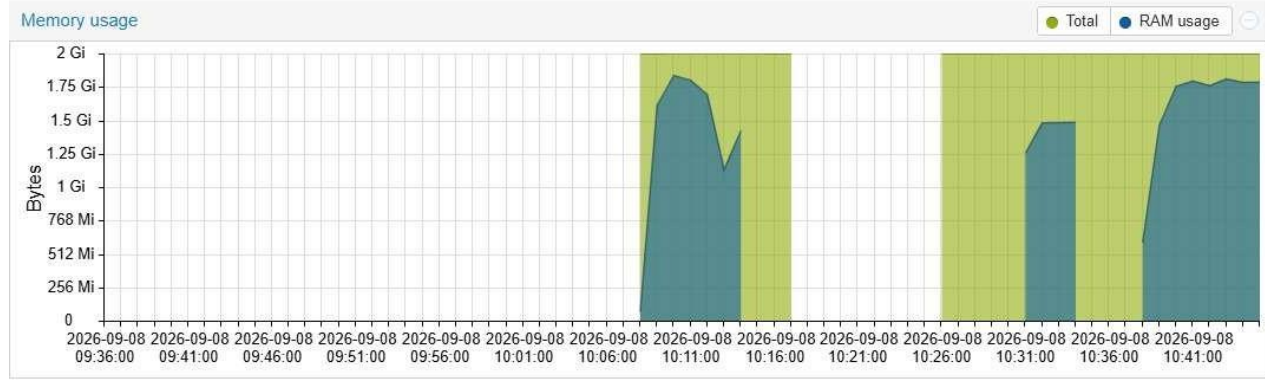


Figure G. Proxmox VE: Memory usage graph over the observed time window.

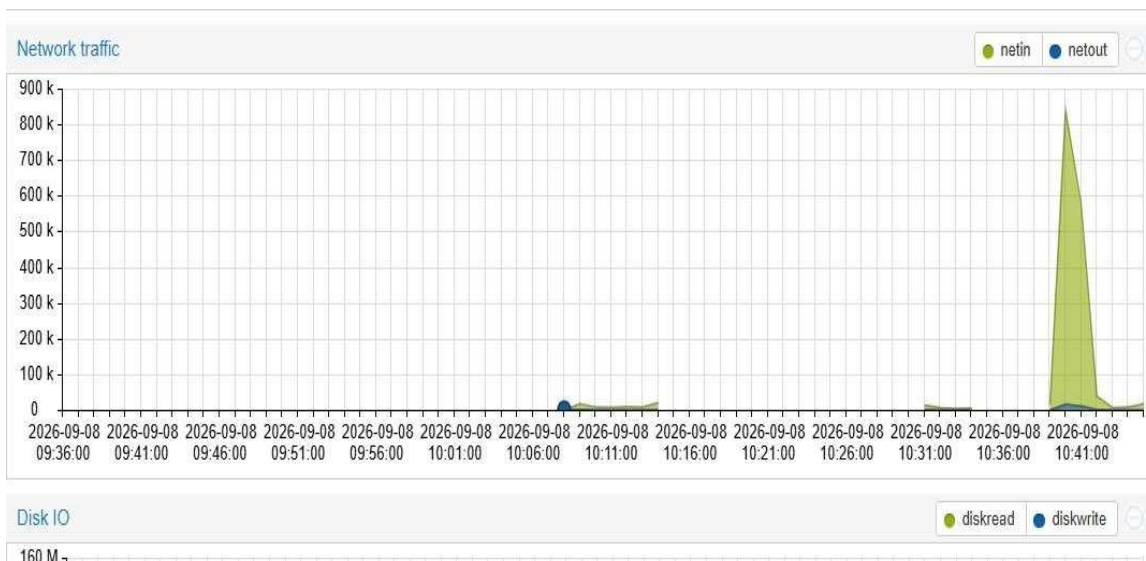


Figure 10. Proxmox VE: Network traffic graph showing the observed network activity and transient spike.

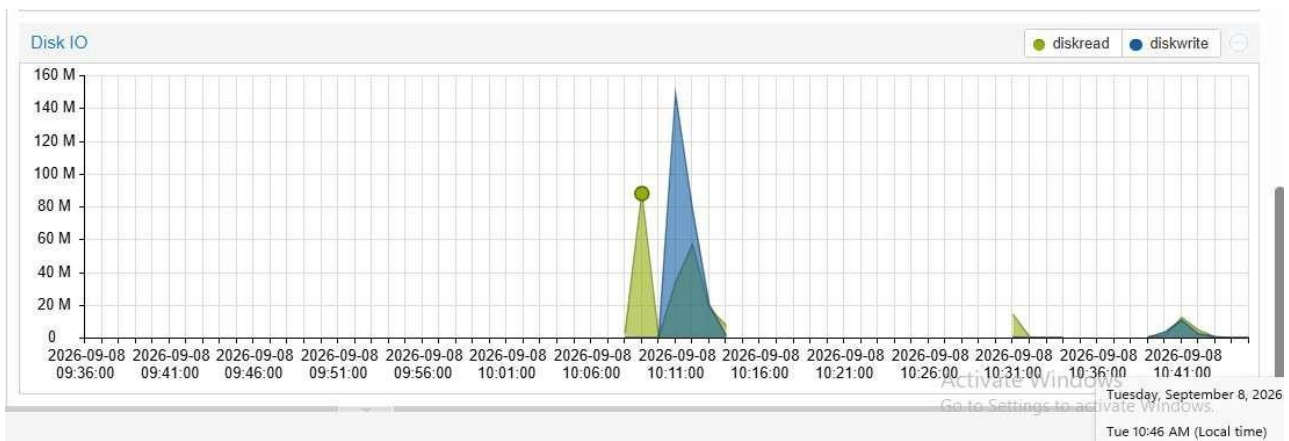


Figure 11. Proxmox VE: Disk I/O graph showing disk read/write activity during the observation period.

9. Final Observation Record

Type-1 Hypervisor Performance Record

Parameter	Part 1 Result
Hypervisor	Proxmox VE
Hypervisor type	Type-1
Guest OS	Ubuntu 22.04.5 LTS
CPU	2 vCPU
Memory	2 GB
Disk	20 GB
Total execution time	10.0005 s
Total events	15,877
Events per second	1,587.47
Average latency	0.63 ms